Xenon reacts with $PtF_6$ and gave an orange yellow solid [$XePtF_6$] and this is insoluble in $CCl_4$.

Xenon difluoride forms addition compounds $XeF_2.2SbF_5$ and $XeF_2.2TaF_5$. Xenon hexa fluorides forms compound with boron and alkali metals. Eg : $XeF_6.BF_3$, $XeF_6MF$, M-alkali metals.

There is some evidence for existence of xenon dichloride $XeCl_2$.

Krypton form krypton difluoride when an electric discharge is passed through Kr and fluorine at 183° C or when gases are irradiated with $SbF_5$ it forms $KrF_2.2SbF_3$.

**Table 3.11 Structures of compounds of Xenon:**

| Compound | Hybridisation | Shape / Structure |
|---|---|---|
| $XeF_2$ | $sp^3d$ | Linear |
| $XeF_4$ | $sp^3d^2$ | Square planar |
| $XeF_6$ | $sp^3d^3$ | Distorted octahedron |
| $XeOF_2$ | $sp^3d$ | T Shaped |
| $XeOF_4$ | $sp^3d^2$ | Square pyramidal |
| $XeO_3$ | $sp^3$ | Pyramidal |

## Uses of noble gases:

The inertness of noble gases is an important feature of their practical uses.

### Helium:

1. Helium and oxygen mixture is used by divers in place of air oxygen mixture. This prevents the painful dangerous condition called bends.
2. Helium is used to provide inert atmosphere in electric arc welding of metals
3. Helium has lowest boiling point hence used in cryogenics (low temperature science).
4. It is much less denser than air and hence used for filling air balloons

### Neon:

Neon is used in advertisement as neon sign and the brilliant red glow is caused by passing electric current through neon gas under low pressure.

### Argon:

Argon prevents the oxidation of hot filament and prolongs the life in filament bulbs

### Krypton:

Krypton is used in fluorescent bulbs, flash bulbs etc...

Lamps filed with krypton are used in airports as approaching lights as they can penetrate through dense fog.



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